Shenghan Luo

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EDUCATION

Shanghai Jiao Tong University (SJTU) & Shanghai AI Laboratory
Doctoral Student in Electronic Information, Joint Training Program | Sept. 2026 - Present
Southeast University (SEU) *Chien-Shiung Wu College, Future Technology Class*
B.Eng in Information Engineering | Aug. 2022 - Jun. 2026

RESEARCH INTERESTS

Wireless Communications: Multimodal-Aided Channel Estimation; ISAC; AI4Comm
Multi-Agent Systems: Game Theory; Agent Privacy and Security; Embodied Collaborative Perception
Video Generation: World Models; Real-Time Live Streaming; Generative Video Communications
Machine Learning Theory: Optimization; Manifold Learning; Statistical Inference

PUBLICATION

Shenghan Luo. "Uncertainty Quantification of Least Squares Estimator for Generalized Orthogonal Procrustes Problems." ***Conference on Neural Information Processing Systems (NeurIPS 2026) (Accepted).***
Shenghan Luo, Jiale Huang, Ziyao Xie, et al. "Transitional Scenario Channel Characteristics Analysis for 6G Wireless Communications." ***IEEE International Conference on Communication Technology (ICCT 2025) (Accepted).***
Wei Jia, Chao Zhu, **Shenghan Luo**, et al. "Swin Transformer Aided Urban Digital Twin Online Channel Modeling Platform." ***IEEE Vehicular Technology Conference (VTC2025-Fall) (Accepted).***
Shenghan Luo, Yin Xu, Jie Yang, Yang Wang, Cixiao Zhang, Shi Jin, and Wenjun Zhang. "SkySense: A Semi-Supervised Generative Framework for UAV Localization in ISAC Networks." ***IEEE Global Communications Conference (GLOBECOM 2026) (Accepted).***

RESEARCH EXPERIENCES

Uncertainty Quantification of Least Squares Estimator for Generalized Orthogonal Procrustes Problems
Jan. 2025 - Present
• Derived the **second-order analytic expansion** and **blockwise asymptotic distribution** of the global least-squares estimator under additive Gaussian noise, accounting for anisotropic point-cloud geometries.
• Formulated the Riemannian Newton step as a **continuous Sylvester equation** to preserve tangent-space constraints and characterize geometry-dependent estimation uncertainty.
• Constructed **data-driven chi-square confidence regions** on the orthogonal group and validated the error expansion, asymptotic distribution, and nominal coverage through numerical experiments.

Analysis and Simulation of Channel Characteristics for Typical Outdoor Centimeter-Wave Scenarios Based on 6G Pervasive Channel Model
National Mobile Communications Research Laboratory, Aug. 2023 - Mar. 2025
Leader of National Student Research Training Program (SRTP) Project*, supervised by Prof. Jie Huang and Prof. Chengxiang Wang*
• Derived dedicated channel model parameters for 12 outdoor scenarios in the **6G Pervasive Channel Model (6GPCM)** within 6GPCS software through detailed analysis of channel measurement data.
• Completed a comprehensive simulation and analysis of channel characteristics for different outdoor environments, including **RMS delay spread, doppler spread and angular spread**, while evaluating the sensitivity of key channel characteristics to environment dynamics.
• Designed a novel simulation scheme for evaluating channel characteristics under **transitional scenarios** between two defined communication scenarios employing **Conditional Generative Adversarial Network** which reduce the dependency on real channel mesurement data.

Real-Time 3D Urban Channel Simulation for Digital Twins
National Mobile Communications Research Laboratory, Aug. 2023 - Mar. 2025
Core member of National SRTP project*, supervised by Prof. Junling Li and Prof. Chengxiang Wang*
• Developed **3D electromagnetic-aware urban models** in Blender by initiating with GIS map data of open source and adjusting detailed building geometries and material textures according to real-world electromagnetic properties, and validated the model accuracy through Wireless Insite simulations.
• Designed and implemented a **multifunctional interactive UI** in Unity enabling real-time 3D channel visualization, dynamic scenario analysis, and cross-environment data comparison through camera perspective controls, customizable heatmaps, and automated scene recognition features.
• Designed and trained novel **Swin Transformer models** for high-precision semantic segmentation of urban remote sensing images from GIS data, enabling automated classification of multiple complex communication scenarios. Integrating results with static models and user interfaces, assigning corresponding channel types to different urban scenes in Unity to simulate channel characteristics using 6GPCM.

SkySense: Semi-Supervised Generative UAV Localization in ISAC Networks
Southeast University (SEU) and Shanghai Jiao Tong University (SJTU)
Undergraduate Thesis*, supervised by Prof. Shi Jin, Assoc. Prof. Jie Yang, and Prof. Yin Xu*
• Developed **SkySense**, a semi-supervised framework for 3D UAV localization in ISAC networks; learned spatial representations from unlabeled CSI sequences using **self-supervised temporal contrastive learning**.
• Designed a **conditional consistency generative model** to address multipath spatial ambiguity with only **1-2 inference steps**, jointly fine-tuning the encoder using a limited set of labeled coordinates.
• Integrated distributed multi-base-station fusion and achieved a **9.77 cm mean 3D localization error** with **1% labeled data** under a **3-BS setup** on a real-world measurement dataset.

AWARDS

• **2022-2023 & 2023-2024** SEU Merit Student (Sanhao Student)
• **Southeast University BYD Scholarship (¥10,000)**
• **2025 Wang Qi Scholarship (¥5,000)**
• **Third Prize of Future Technology College Taihu Scholarship (¥5,000)**
• **Southeast University Zhishan Scholarship (¥2,000)**
• **Second Prize** of the 21st Jiangsu Provincial College Advanced Mathematics Competition
• **Second Prize** of 2024 Jiangsu Provincial College Students' Electronic Design Contest

ACADEMIC SERVICE

Reviewer: International Conference on Artificial Intelligence and Statistics (AISTATS 2026); Conference on Neural Information Processing Systems (NeurIPS 2026); IEEE Transactions on Communications (TCOM).

EXTRACURRICULAR ACTIVITIES

• 2022-2025 Core Member, Psychology Association, Student Union of Chien-Shiung Wu College, Southeast University
• Peer mentor of Class 61524G of Chien-Shiung Wu College, SEU
• 2023-2024 Head of the "Advanced Mathematics" group in the school's Q&A room
• Accumulated volunteer hours: **164.8 hours**, rated as a **three-star volunteer**

SKILLS

Programming Languages: C, C++, Python, MATLAB, Verilog, VHDL, LaTeX
Specialized Software: Vivado, Quartus II, FDTD, Blender, Wireless Insite, Sionna | **Hardware:** FPGA, STM32